FIDO2 CTAP 2.3 & WebAuthn L3 | ML-DSA-87 (COSE alg: -50) | RFC 8554 LM-OTS (w=4, p=67) | Google Longfellow ZK

CTAP 2.3 Wire Size Analysis: Attestation over an ML-DSA-87 Credential

Because the authenticator signs authData || clientDataHash using LM-OTS (w = 4, p = 67), the attestation statement (attStmt) size itself does not change at all when upgrading the credential from ML-DSA-65 (alg: -49, 1,952 B public key) **to ML-DSA-87 (alg: -50, 2,592 B public key). Only authData grows by +640 bytes.**

1. authData (0x02) Breakdown with an ML-DSA-87 Credential: 2,689 Bytes

authData Component	Byte Size	Details
`rpIdHash`	`32 B`	SHA-256 of Relying Party ID
`flags`	`1 B`	UP, UV, AT=1, ED
`signCount`	`4 B`	32-bit signature counter
`aaguid`	`16 B`	Authenticator model identifier (16 bytes)
`credentialIdLength + credentialId`	`34 B`	2 B length + 32 B credential ID (or 18 B if using a 16-byte ID)
credentialPublicKey (COSE_Key)	`2,602 B`	10 B canonical CBOR COSE_Key (kty: 7, alg: -50) + 2,592 B ML-DSA-87 PK
Total authData Payload	**`2,689 B`**	**Plus 4 B outer CBOR key/length header = 2,693 B on wire**

In-circuit SHA-256 impact: authData (2,689 B) || clientDataHash (32 B) + 52 B LM-OTS header = 2,773 bytes = 44 SHA-256 blocks (up by just +11 blocks from ML-DSA-65's 33 blocks).

2. Authenticator Attestation Statement (attStmt, 0x03) Breakdown

The attStmt emitted by the authenticator over the CTAP 2.3 bus is identical in size for both ML-DSA-65 and ML-DSA-87:

attStmt Field	Consumer 31-Level (14 Nodes)	Consumer 31-Level (All 31)	Enterprise 25-Level (No ZK)
LM-OTS Signature (w=4, p=67)	**2,180 B** (4B type + 32B C + 67x32B)	`2,180 B`	`2,180 B`
Merkle Path Nodes	**448 B** (14 x 32 B device nodes)	**992 B** (31 x 32 B full path)	**800 B** (25 x 32 B full path)
Routing & Domain IDs	**68 B** (batchRootId 32B + q 4B + l 32B)	`68 B`	`68 B`
CBOR Map Framing (fmt+attStmt)	`~44 B`	`~44 B`	`~44 B`
Total attStmt + fmt on Wire	**`2,740 B (~2.68 KB)`**	**`3,284 B (~3.21 KB)`**	**`3,092 B (~3.02 KB)`**

3. Total CTAP 2.3 Wire Size (authData + attStmt + fmt) vs. the 7,609 B USB HID Limit

Deployment Mode (ML-DSA-87)	authData	attStmt + fmt	Total Wire Response	Headroom (<7,609 B)
1. Consumer 31-Level (14 Dynamic Nodes from Key)	`2,693 B`	`2,740 B`	5,433 B (~5.31 KB)	**2,176 B to spare** (92 HID frames)
2. Consumer 31-Level Self-Contained (All 31 Nodes from Key)	`2,693 B`	`3,284 B`	5,977 B (~5.84 KB)	**1,632 B to spare** (101 HID frames)
3. Enterprise Direct 25-Level (All 25 Nodes from Key, No ZK)	`2,693 B`	`3,092 B`	5,785 B (~5.65 KB)	**1,824 B to spare** (98 HID frames)

Key Takeaway

- The attestation portion itself (attStmt) is ~2.68 KB to ~3.21 KB on the wire, and the complete CTAP 2.3 **makeCredential response including the 2,592 B ML-DSA-87 public key is ~5.31 KB to ~5.84 KB.**
- Every deployment mode fits comfortably inside a single 7,609-byte USB CTAPHID message with 1.6 KB to 2.1 KB of headroom to spare (whereas standard X.509 'packed' attestation with ML-DSA-87 requires ~13-18 KB and overflows).